

L2A™

(c) Discuss the concept of 'Mendelian population' and its application in study of anthropo-genetic variation in India

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The concept of Mendelian Population was introduced by Sewall Wright in his "Evolution in Mendelian Population (1931)". He defined it as:

"An interbreeding population that shares a common gene pool".

Others such as Mather & Gregg, Dobzhansky have said it is: "Spacio-Temporal group of conspecific interbreeding individuals".

<Characteristics>

- Genetically closed system & forms a 'breed isolate'.
- The members share in 'gene pool'.
- Matings within group are random & fertile & usually likely.
- Cutoff from outsiders. It might be due to geography, social

Values, beliefs, practices, etc. - L2A™

- exists in a hierarchy, infact a Mendelian population can contain several mendelian populations (M.P) with in it also.
- largest possible Mendelian Popⁿ is the species itself

Ex: Caste system of India in Jati [kin-Group] Model as means of marriage allians Charactered by Endogamy [Mating within defined social unit]

<Application>

- we can study effect of Inbreeding and find out the genetic structure and evolution of population.
- When paired with Hardy weinberg law we find a deviation. So we can find what is the cause of deviation which is resulting in Evolution. Since

evolution is change in the gene frequencies and since **L2A™** M.P. is inbreeding it also is said to be evolving.

- The spread of traits can also be found out. Also used in the Peoples of India Project, 1985 by KS Singh of Anthropological Survey of India also.

- helps understand population boom in India and tells us what steps to be taken by the Health Mins to control it.

Thus M.P. have huge role in the field of Population Growth especially when Hybridisation and Migration are increasing.